

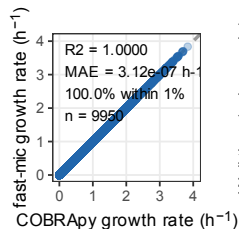
A

The fast-mic pipeline



B

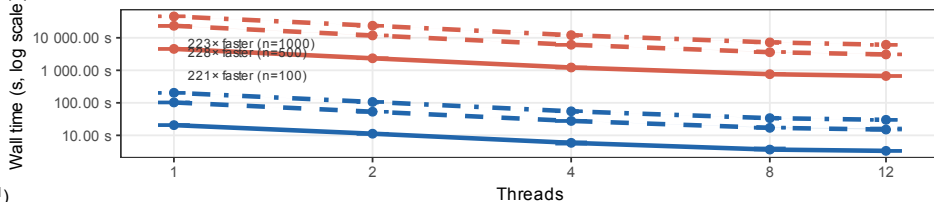
Numerical agreement



C

Runtime

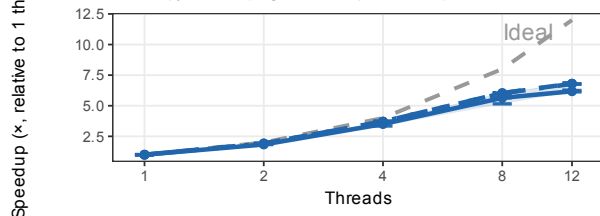
fast-mic vs COBRApy at matched core counts



D

fast-mic parallel speedup

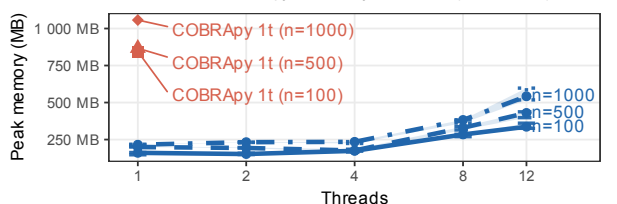
COBRApy omitted (single-threaded per FBA call)



E

Memory usage

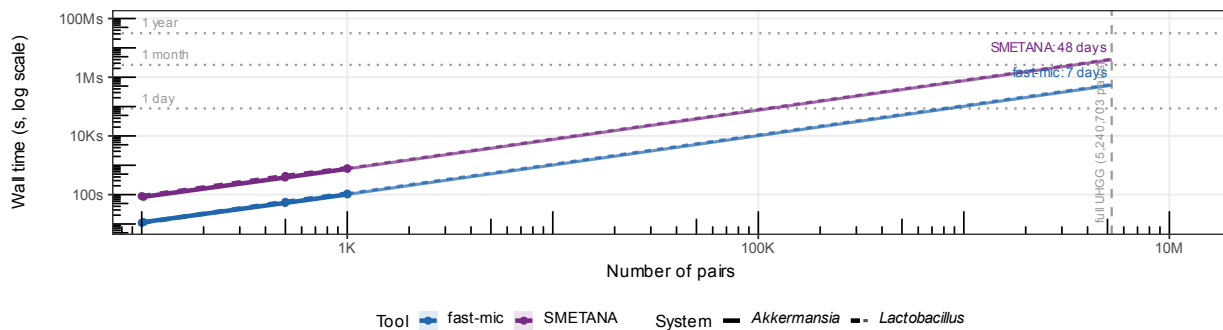
fast-mic curve; COBRApy shown only at 1 thread (true baseline)



F

Interaction-typing scalability

fast-mic vs SMETANA (single-thread); dashed = projection to full catalogue



G

Memory cost of interaction typing

same pairs, single process, single thread; grey = SMETANA/fast-mic ratio

